FACULTY OF ENGINEERING AND TECHNOLOGY

FINAL ASSESSMENT BIS3106 PROJECT MANAGEMENT

GROUP PROJECT

ACADEMIC SESSION: SEPTEMBER 2025

RELEASE DATE: 18 MAY2026, FRIDAY

SUBMISSION DEADLINE: 17 JULY 2026, FRIDAY

INSTRUCTIONS TO CANDIDATES

- This assignment will contribute **50% to your final grade.**
- This is **a group project**. Each group **consists of ONLY 5 members**.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as non-submission and marked zero.

Students' declaration:

We;

	(Name)	(ID)	(Signature)
1.	Vinnci Chin Wen Xi	22057269	Vinnci Chin
2.	Liang Yrenn Da	22067128	liang
3.	Ethan Soong Ren Wei	25004565	***Ethan***
4.	Yeoh Jie Long	22071112	Jie Long
5.	Ooi Jin Yon	22039069	Jin Yon

received the assignment and read the comments.

Academic Honesty Acknowledgement

"We (names stated above) verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to page 1C, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

1) Vinnci Chin Wen Xi Vinnci Chin / 29-6-2026
2) Liang Yrenn Da liang / 29-6-2026
3) Ethan Soong Ren Wei ***Ethan*** / 29-6-2026
4) Yeoh Jie Long Jie Long / 29-6-2026
5) Ooi Jin Yon Jin Yon / 29-6-2026

Project Name:	**Student Collaboration Management System**
Project Manager:	Vinnci Chin Wen Xi

0. PROJECT SUMMARY

The Student Collaboration Management System (SCMS) is a web-based platform designed to assist university students in forming effective assignment and project teams through intelligent teammate matching. Based on academic studies, skills, availability and preferred working style, communication preferences and project roles, the system suggests the most appropriate team members. SCMS helps address some of the typical collaboration problems that occur when forming a team, including time conflicts, unequal workloads, and conflicting working styles. The platform facilitates better communication, fairer task distribution and a more efficient group project experience. This project involves planning and designing a proposed system that will include a project scope, requirements, resources, and implementation strategies. We are developing a platform that helps students find compatible teammates for assignments and projects based on course, skills, confidence level, preferred working style, availability, communication preference, and leadership preference. The system functions similarly to a professional networking and matching platform for academic teamwork.

1. PROJECT BACKGROUND

Group assignments are common in universities, but students frequently experience difficulties such as:

- Inactive team members
- Unequal workload distribution
- Poor communication
- Schedule conflicts
- The dissatisfaction over random group formation.
- Difficulty finding suitable teammates

The issues have a negative impact on the quality of teamwork, project results and student satisfaction. Currently, most students are grouped by personal network or random grouping, not taking into account skill compatibility and availability. The Student Collaboration Management System solves this problem by offering an intelligent matching platform to suggest appropriate teammates and help with project collaboration.

2. PROJECT OBJECTIVES

Objective 1

To develop a student profile management feature that allows students to record their skills, availability, communication preferences, and preferred project roles.

Objective 2

Make intelligent teammate suggestions based on teammate skills, schedules and working preferences.

Objective 3

Allow group collaboration to be efficient and manage group project conflicts by providing group management, task tracking and communication tools

3. PROJECT DETAIL

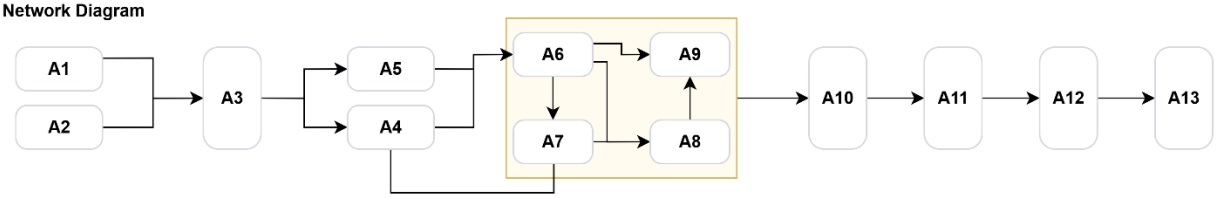
3.1 Project Scope

Project Tasks, Processes, Workflow and Network Diagram

NO	ACTIVITY	DEPENDENCY	DESCRIPTION	EXPLANATION
A1	Requirement analysis	-	Identify user needs and system requirements	First step in understanding the requirements of a project
A2	Research existing platforms	-	Analyse similar platforms and identify useful features and limitations.	Can be conducted together with requirement gathering.
A3	Define system features	A1, A2	Determine core functions	Features depend on user needs and

				research findings.
A4	Design system architecture	A3	Plan system structure, database tables, user data storage, and relationships between system components.	Confirmed features are required before developing a system structure.
A5	Design UI prototype	A3	Create prototypes for registration, profile setup, matching results and group management pages.	Finalised functions are a prerequisite to designing an interface.
A6	Develop user profile module	A4, A5	Build features allowing students to create accounts and enter skills, courses, availability, and preferences.	Database structure and interface design are needed for development.
A7	Develop matching algorithm	A4, A6	Build matching logic to suggest suitable teammates based on compatibility.	Student profile data must be stored to match.
A8	Develop group management features	A6, A7	Develop group creation, member management, role assignment and participation tracking functions.	Users and matching are required before groups can be created.
A9	Develop communication features	A6, A8	Build tools for team interaction such as messaging, scheduling, announcements or discussion areas.	Created groups and users are required for communication.
A10	Integrate system modules	A6, A7, A8, A9	Combine all developed features into one functional web-based platform.	All modules must be completed before integration.
A11	Perform system testing	A10	Conduct functionality testing, usability testing, and fix system errors.	A complete integrated system is required for testing.
A12	Conduct user evaluation	A11	Gather feedback from the University students and assess the effectiveness of the system.	Users test once system issues have been resolved.
A13	Deployment	A12	Release the completed platform for student use.	Once the evaluation is successful, it proceeds to deployment.

Network Diagram



3.2 In scope and out of scope

In Scope	
1	Student registration and login system.
2	Student profile creation and management.
3	Skill, availability, and working preference management.
4	Intelligent teammate matching based on predefined compatibility criteria.
5	Group formation through invitations and request management.
6	Team dashboard for project monitoring and task tracking.
7	Contribution tracking and participation monitoring.
8	Peer evaluation feature for assessing teammate contributions.
9	User interface (UI) prototype and system design documentation.
10	Database design and system architecture planning.

Out of Scope	
1	Real-time video conferencing and online meeting features.
2	Social networking functions unrelated to academic collaboration.
3	AI-based academic performance prediction.
4	Third-party authentication services (e.g., Google Login, Microsoft Login).
5	Development of a fully operational system beyond the prototype and planning stage.
6	Financial transactions or payment-related functions.
7	Commercial deployment and production hosting.
8	Support for external users outside the university community.

3.3 Timeline and Milestones

Gantt Chart				Week													
Milestone		Activities and Task		1	2	3	4	5	6	7	8	9	10	11	12	13	14
M1	Requirements Analysis	A1	Requirement analysis														
		A2	Research existing platforms														
M2	System Design	A3	Define system features														
		A4	Design system architecture														
		A5	Design UI prototype														
M3	Core System Development	A6	Develop user profile module														
		A7	Develop matching algorithm														
		A8	Develop group management features														
		A9	Develop communication features														
M4	System Testing & Refinement	A10	Integrate system modules														
		A11	Perform system testing														
M5	Project Deployment	A12	Conduct user evaluation														
		A13	Deployment														

3.4 Key Deliverables

TASK	DEPENDENCIES	DELIVERABLES	ACCEPTANCE CRITERIA
Requirement Analysis & System Design **(A1–A4)**	A1, A2	System Requirements and Architecture Document	Core functions such as matching profiles are defined. System structure and database tables are fully planned.
User Interface Design **(A5)**	A3	UI Prototype / Wireframes	Registration, profile setup, matching results and group pages are accepted as wireframes.
Core Modules Development **(A6–A9)**	A4, A5	Functional System Modules	Student profile creation, intelligent matching algorithm, and group management features are built and working.
System Integration & Testing **(A10–A11)**	A6, A7, A8, A9	Integrated Web-Based Platform	All modules are combined into a system. Functionality and usability testing are complete with errors fixed.
User Evaluation & Deployment **(A12–A13)**	A10, A11, A12	User Feedback Report & Final Released Prototype	Feedback is collected from university students and the prototype is successfully deployed for student use.

Exclusions, Assumptions and Constraints	
Exclusion	Real-time video conferencing, third-party authentication, integration with existing university systems, and commercial deployment.
Assumptions	Students will have stable internet access and the necessary devices to access the web-based platform.
Constraints	The project must be completed and submitted by the strict deadline of 17 July 2026. The project team is strictly limited to 5 members.

3.5 Project Budget

COST TYPE	ITEM	RATE (RM)	QTY	AMOUNT (RM)
Labour	Project Manager	600.00	1	600.00
Labour	System Analyst	500.00	1	500.00
Labour	UI/UX Designer	500.00	1	500.00
Labour	Front-end & Back-end Developers	700.00	2	1,400.00
Labour	QA/Test Engineer	500.00	1	500.00
Software	Figma Professional Subscription	75.00	1	75.00
Software	Domain Name Registration	60.00	1	60.00
Software	Cloud Hosting & Database Services	180.00	1	180.00
Software	GitHub Pro / Version Control Tools	50.00	1	50.00
Testing	User Acceptance Testing Materials	100.00	1	100.00
Miscellaneous	Documentation & Printings	80.00	1	80.00
Contingency	Project Contingency Reserve (10%)	405.00	1	404.50
TOTAL ESTIMATED COST				**RM4,449.50**

The total estimated project budget includes all the necessary resources to plan, design, develop, test, deploy, and document the Student Collaboration Management System. This project is not intended for commercial production, therefore the estimated costs reflect project effort, critical software subscriptions, cloud based service costs, testing activities and a contingency reserve for the unforeseen. The budget offers a realistic and feasible financial estimate and should enable the project's objectives to be achieved within the timeframe set.

3.6 Project Resource

Resource Category	Resource	Quantity	Responsibilities / Purpose
Project Management	Project Manager	1	Responsible for managing the project and coordinating all its activities, including planning, scheduling, allocating resources, monitoring risks, communicating with stakeholders, tracking progress, and ensuring overall project coordination.
Business Analyst	System Analyst	1	Collect and examine user needs, evaluate current collaboration tools, create functional specifications, verify project needs, and assist in scope definition.
UI/UX Design	UI/UX Designer	1	Design user interface wireframes, develop user experience flows, create interactive prototypes, and ensure usability and accessibility requirements are met.
Software Development	Front-end Developer	1	Develop web interface, registration module, user interface management, dashboard, and responsive user interface elements.
	Back-end Developer	1	Implement business logic, player pairing algorithm, API, authentication, messaging capabilities, and system integration.
Database Management	Database Engineer	1	Develop database schemas, create database structure, optimize queries, ensure database consistency, and enable integration with various systems.
Quality Assurance	QA / Test Engineer	1	Perform functional testing, usability testing, integration testing, bug reporting, regression testing, and verify that project requirements have been achieved.
End Users	University Students (UAT Participants)	20–30	Implement User Acceptance Testing (UAT), test system usability, and give feedback for improvements prior to deployment.
Hardware	Developer Laptops	5	Used in software development, UI design, documentation, database implementation, testing, project management activities.
Software Tools	Visual Studio Code, MySQL, GitHub, Figma, Microsoft Project	5 Licences / Shared Tools	Support software development, UI/UX design, version control, database management, documentation, project scheduling and collaboration.
Cloud Infrastructure	Web Hosting & Cloud Database Server	1	Host prototype system, offer database services, and facilitate the deployment and demonstration of the finished web-based platform.

Project resources are assigned based on the various stages of the project to ensure the availability of personnel, hardware, software, and infrastructure as and when needed. This allocation reflects the efficient use of resources, is consistent with the work scope, key deliverables and project schedule, and will help to deliver the Student Collaboration Management System within the allocated budget and timeframe.

3.7 Project Management Tools

The PM tools used in this project include a stakeholder register, Gantt chart, and a risk assessment matrix in developing the schedule, stakeholder, and risk performance domains within the process groups of pre-initiating, initiating, planning, executing, monitoring and controlling, and closing a project’s life cycle through the use of Microsoft Project to ensure that each phase is systemically followed based on a determined structure and methodoloy.

4. PROJECT TEAM AND STAKEHOLDERS

4.1 Project Team

NO.	NAME	ROLE	Signature
1	VINNCI CHIN WEN XI	Project Manager	Vinnci Chin
2	LIANG YRENN DA	Frontend Developer	liang
3	YEOH JIE LONG	Backend Developer	Jie Long
4	OOI JIN YON	UI/UX Designer	Jin Yon
5	ETHAN SOONG REN WEI	Tester	Ethan

4.2 Stakeholders

Stakeholder Register				
Name	**Role**	**Power**	**Interest**	**Contact Information**
Sunway University & Academic Board	Project Funder / Regulatory Body	High	High	N/A
Vinnci Chin Wen Xi	Project Manager	High	High	22057269@imail.sunway.edu.my
Ooi Jin Yon, Liang Yrenn Da, Yeoh Jie Long, Ethan Soong Ren Wei	Project Team Members	Medium	High	22039069@imail.sunway.edu.my 22067128@imail.sunway.edu.my 22071112@imail.sunway.edu.my 25004565@imail.sunway.edu.my
Sunway University Students & All University	End Users/ Target Market	Medium	High	N/A
System developers	Frontend Developer, Backend Developer and UI/UX Designer	High	High	*See Project Team Members*

Stakeholder Map			
Level of Influence	**Keep Satisfied**		**Manage Closely**
High			Sunway University & Academic Board
			Project Team Members
			Students & All University
Medium			Project Team Members
Low			
	Monitor		**Keep Informed**
Level of Interest	**Low**	**Medium**	**High**

5. PROJECT RISKS

RISK MATRIX		SEVERITY (IMPACT)				
		INSIGNIFICANT	MINOR	MODERATE	MAJOR	SEVERE
LIKELIHOOD	ALMOST CERTAIN	Low Medium	Medium	Medium High	High	High
	LIKELY	Low	Low Medium	Medium	Medium High	High
	POSSIBLE	Low	Low Medium	Medium	Medium High	Medium High
	UNLIKELY	Low	Low Medium	Low Medium	Medium	Medium High
	RARE	Low	Low	Low Medium	Medium	Medium

RISK	LIKELIHOOD	IMPACT	RESPONSE	RISK LEVEL	RISK OWNER
1: Students provide inaccurate or misleading profile information	Possible	Moderate	Implement profile verification, lecturer review, and validation checks.	Medium	System administrator
2: Privacy and security breach of student data	Likely	Major	Implement secure login, encryption, access control, and privacy settings.	Medium High	Development Team
3: System downtime and recovery maintenance	Likely	Severe	Implement downtime maintenance procedures and backup controls	High	System administrator
4: Usability satisfaction of the UI/UX of the system in stakeholder's perspective	Possible	Minor	Re-negotiate requirements or implement backup systems in catering to the different user satisfactions	Low Medium	System UI/UX developer
5: Misuse of system features (bias towards selecting the same student choice)	Possible	High	Perform algorithm analysis to ensure proper randomisation of student choices to ensure the same student cannot be chosen multiple times at once.	Medium High	Sunway University Students
6: Mismanagement or poor communications between project manager and development teams	Possible	Severe	Implement risk response analyses and perform management control procedures	Medium High	Project Manager

6. EXPECTED OUTCOMES AND SUSTAINABILITY IMPACT

Expected Outcomes:

Our system aims to provide outcomes that assists in the domains of:

- Increases student satisfaction in ensuring that qualified teammates can bring out their full potential.
- Increases teamwork productivity by bringing together qualified teammates.
- Reduces scheduling conflicts to ensure timely submission and collaboration.
- Reduces the risk of wasting time and resources when searching for group members with the right skills.
- Reduces the risk of potential free riders in group projects or assignments.
- Minimizes the outgoing risk of missing out isolated students that have high potential.

Sustainability Impact Mapping:

Our project achieves a sustainability goal such as SDG 4: Quality Education. It satisfies the following sustainability impacts:

- Ensures that project team matchmaking assists in bringing out a team's full potential without the high possibility of snoozing teammates hindering the effectiveness of a group's work productivity when working on assignments together.
- Ensures quality education by prioritising student satisfaction in finding group member skills that match their desired needs.

7. ASSIGNMENT LOG

Assignment Tracker Sheet

Total Assmt.	7	Completed	7	% Complete	100%

Task Breakdown	Subject/ Course	Status	Due Date	Priority Level	Days Left	Remarks
0. Project Summary	BIS3106	Completed	17-Jul-26	Low	3	
1. Project Background	BIS3106	Completed	17/7/2026	Low	3	
2. Project Objectives	BIS3106	Completed	17/7/2026	Medium	3	
3. Project Details	BIS3106	Completed	17/7/2026	High	3	Add PM Tools
4. Project Stakeholders	BIS3106	Completed	17/7/2026	Medium	3	Add Project Team
5. Project Risks	BIS3106	Completed	17/7/2026	High	3	Map Risk Matrix
6. Expected Outcomes	BIS3106	Completed	17/7/2026	Low	3	Map to SDGs